

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mechanical Engineering Dept

2. Subject Name: Computer Controlled Manufacturing Machines

L-T-P: 4-0-0 Credits:4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus: Introduction to levels of production and applicability of computer control in case of machine tools.
Introduction to computer controlled machines - classification: Point to point and continuous control machines, open loop and closed loop control, machining centres and turning centres. Digital logic, devices used in computer controlled machines, Design features of CNC machines, prime movers and their control, feedback control. G & M Code programming, Computer aided Offline programming
Interpolation: software and hardware interpolators Free form surface modeling and machining on 3 axis CNC machines using ball ended milling cutter.
Introduction to Robotics, Intelligent Robots, Sensors for industrial robots, encoders, resolvers, Hall-effect sensors, acoustic sensors, ultrasonic and optical/infrared sensors, Elements of robot vision, Integration using PLCs. Digital motion planning systems

Reference Books:

- 1) Fundamentals of Robotics by D.K. Pratihar, Narosa Publishing House (for the Robotics part marked in RED)
- 2) Yoram Koren, Computer control of manufacturing Systems
- 3) Leach, Malvino and Saha, Digital principles and applications
- 4) Mathematical elements in Computer graphics: Rogers and Adams

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to levels of production and applicability of computer control in case of machine tools.	2
2.	Introduction to computer controlled machines - classification: Point to point and continuous control machines, open loop and closed loop control, machining centres and turning centres.	6

3.	Digital logic, devices used in computer controlled machines, Design features of CNC machines, prime movers and their control, feedback control	6
4.	G & M Code programming, Computer aided Offline programming Interpolation: software and hardware interpolators Free form surface modeling and machining on 3 axis CNC machines using ball ended milling cutter.	8
5.	Introduction to Robotics	8
6.	Intelligent Robots	1
7.	Sensors for industrial robots, encoders, resolvers, Hall-effect sensors, acoustic sensors, ultrasonic and optical/infrared sensors	5
8.	Elements of robot vision	2
9.	Integration using PLCs	1
10.	Digital motion planning systems	5
Total number of hours		